ZOO603-Principles of Fish Biology

Current Midterm Subjective 2023 File by

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1. **Cannibalism**:

Cannibalism is the act of consuming another individual of the same species as food.

2. Detritovores:

• Fish consume decaying plants and animals.

3. Pufferfish poisoning/Fugu poisoning:

• Tetrodotoxin poisoning occurs following the ingestion of pufferfish, boxfish or porcupine fish.

• This is also known as puffer or **fugu poisoning**.

• The poisoning is due to toxin known as tetrodotoxin which is concentrated in the internal organs such as liver, ovary and intestines.

• The skin also contains some glands which secrete these toxins.

• Usually the body muscles are free of the toxin.

4. Lateral line system:

Lateral line system consists of 100 or more sensory organs(neuromasts) that are typically arranged in lines on or just under the skin of head and body. It is absent from all reptiles, birds, mammals even those that are aquatic (such as turtles, dolphins and whales).

5. Chondrostean.

• Chondrostean are thought to be the most primitive of the ray-finned fishes.

• They were at their greatest diversity during the Paleozoic and Mesozoic eras.

• Ancestral chondrostean had a bony skeleton, but living members, the sturgeons and paddlefishes, have cartilaginous skeletons.

• Chondrostean also have a tail with a large upper lobe.

6. Bottom feeders (dwellers):

• Bottom feeders (dwellers) are fish that inhabit the lower level of the aquarium o feed off the bottom. • Their mouth may be turned down or “under slung”

• And they may have barbs to help them locate food. They eat bottom food such as rotten food.
Example:

• Barbs, Marigal, Labeo calbasu, Cyprinus carpio, Cirrhina cirrhosus.

• Some fish do not show a preference for the level of the aquarium.

• **Example**: • • Goldfish

7. Muller's classification (1844):

• Earliest classification was proposed by J. Muller (1844) who proposed six subclasses in class Pieces.

• Class PISCES Subclasses: DIPNOI (Lung fishes); TELEOSTEI (Bony fishes); GANOIDEI (Polypterus, Amia, Lepidosteus); ELASMOBRANCHI (Cartilagenous fishes); MARSIPOBRANCHI (Cyclostomes); LEPTOCARDII (Amphioxus)

8. What kind of food do herbivorous fish eat?

Herbivorous Fish exclusively consume plant matters as food.

This type of fish eats aquatic plants, flowers, fruits, seeds, leaves and young stems. They also eat algae that grow on rocks or pebbles by mouth using wide lips under their snout.

~~8.~~9.General characteristics of Fish:

➢ Fish are aquatic, gill-bearing animals

➢ that lack limbs with digits.

➢ Most fish are cold-blooded,

➢ allowing their body temperatures to

➢ vary as ambient temperatures change.

~~9.~~10. Class cephalospidomorph:

• Sucking mouth

• Teeth are present

• Rasping tongue

• Seven pairs of slits

• Blind olfactory sacs

• Cephalospidomorph were, like most contemporary fishes, very well armored.

• They also lacked a swim bladder, and would not have been able to keep afloat without actively swimming.

• The head shield provided some lift though, and would have made the cephalospidomorph better swimmers than most of their contemporaries.

• The head shield was particularly well developed, protecting the head, gills and the anterior section of the viscera.

Example • Lampreys

~~10.~~11. Systematics and Taxonomy:

Systematics is the study of the diversity of fish and the evolutionary relationships between these organisms

• **Taxonomy** is the scientific study of naming, defining and classifying groups of biological organisms based on shared characteristics

Importance of Taxonomy and systematics

➢ Sound systematic data are important for conservation. Without accurate taxonomy,

➢ it is impossible to identify the species and evaluate their conservation status,

➢ it is impossible to properly manage their fisheries,

➢ it is impossible to evaluate

~~11.~~12. Classification of fish on the basis of feeding habitats

• Basic eating groups among fish are:

- **Carnivores**: Fish exclusively consume animal matters as food.
- **Herbivores**: Fish exclusively consume plant matters as food.
- **Omnivores**: Fish consume both plant and animal matters as food.
- **Larvivores**: Fish exclusively consume insect larvae as food.
- **Planktivorous**: Fish exclusively consume plankton as food.
- **Voracious**: Fish consuming or craving large quantities of food.
- **Detritovores**: Fish consume decaying plants and animals.
- **Cannibalistic**: Cannibalism is the act of consuming another individual of the same species as food.
- **Piscivores**: A piscivore is a carnivorous animal that eats primarily fish.

~~12.~~13. Difference between Actinopterygii and Sarcopterygii.

Subclass Actinopterygii: • (Ray-finned fish)

• Their fins lack muscular lobes that is why called as ray-finned fishes.

• They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy.

• Paired fins supported by dermal rays

• Base of fins is not muscular

• Tail fins almost equal to upper and lower lobes

• Blind olfactory sacs

Examples • Ray fined fishes

Subclass Sarcopterygii: • (Lobe-finned fish)

• Have muscular lobes associated with their fins and usually use lungs for gas exchange.

• Pneumatic sacs function as lungs or swim bladder

• Paired fins with muscular lobes

• Some bony fishes are hermaphrodites and a number of species exhibit parthenogenesis

• Fertilization is usually external

• Development is usually oviparous (egg laying) but can be ovoviviparous or viviparous

• Examples • Lungfishes, coelacanths

~~13.~~14. Types of scales in fish Scales in fish

• Different fishes have different types of scales.

• These different types of scales are made of different types of tissue.

• Types of scales also correspond to evolutionary relationships.

There are four main kinds of scales and numerous variations of each kind.

• **Placoid** (sharks and rays)

• **Cosmoid** (lungfishes and some fossil fishes)

• **Ganoid** (bichirs, Bowfin, paddlefishes, gars, sturgeons)

• **Cycloid** and **Ctenoid** (most bony fishes)

- **Placoid scales** are found in the sharks and rays
  - Placoid scales are made of a flattened base with a spine protruding towards the rear of the fish.

- These scales are often called dermal denticles because they are made from dentin and enamel, which is similar to the material teeth are made of.

➢ **Ganoid scales** are flat and do not overlap very much on the body of the fish.

- They are found on gars and paddlefishes.
- In the sturgeon, ganoid scales are modified into body plates called scutes.

➢ **Cycloid scales**

- Cycloid are found in the vast majority of bony fishes.
- These types of scales can overlap like shingles on a roof, which gives more flexibility to the fish.
- These scales also form growth rings like trees that can be used for determining age

➢ **Ctenoid scales**

- Ctenoid scales are different than cycloid scales in that cycloid scales tend to be more oval in shape.
- Ctenoid scales are more clam shaped and have spines over one edge.
- Cycloid scales are found on fishes such as eels, goldfish, and trout.
- Ctenoid scales are found on fishes like perches, wrasses, and parrotfish. Some flatfishes, like flounder, have both cycloid and ctenoid scales.

➢ **Cosmoid Scales:**

Beside the scale types, there are also Cosmoid scales, as well scaleless fishes (many catfish, sculpins, some eels and swordfish) and fishes which have scales so deeply buried that they look scaleless (many tunas and anguillid).